

# Waitlist COLA: A Record-Blind NBA Draft Order With No Tanking Reward

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

## Introduction

National Basketball Association (NBA) teams that fall out of the playoff race often stop trying to win, because finishing worse earns a better draft pick. Roughly a third of the league's teams were openly doing this in the 2025-26 season. The league has answered with ten years of odds adjustments, most recently the sixteen-team 3-2-1 format adopted in May 2026. Banchio and Munro (SSAC 2020) proved that odds alone cannot fix it, because any rule giving worse records better odds still rewards losing.

Carry-Over Lottery Allocation (COLA) gives each team a waiting number that rises every year it misses the playoffs and falls when it succeeds, and sets draft priority by that number, so one bad season barely moves a team. It has many versions. Which one ends the reward for losing, and where in a draft does that reward live?

## Methods

We describe COLA's versions with seven settings and run nine draft rules through ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move. Each rule ran 48 separate leagues of 15 seasons, 14,004 league-seasons in all. Every rule faces three tests. Does losing pay? Does help reach the longest-waiting teams? Can a team game the pool another way?

Six of the nine rules are COLA versions. One is new. Waitlist COLA keeps the four-pick televised draw exactly as the original runs it, then fills picks 5 through 14 by waiting number, longest wait first, ties broken at random.

## Results

Most of the reward for losing came from picks 5 through 14. Table 1 switches each part of the old rule off in turn. Making the draw ignore record while leaving those ten alone raises the reward from 1.30 draft places to 1.98. Making the ten ignore record instead cuts it to 0.61.

| What the rule reads                        | Reward for losing<br><i>draft places</i> | Vs. 2026 rule |
|--------------------------------------------|------------------------------------------|---------------|
| Both parts read record (rule through 2026) | 1.30                                     | baseline      |
| Draw ignores record, later picks by record | 1.98                                     | 1.52×         |
| Later picks ignore record, draw unchanged  | 0.61                                     | 0.47×         |
| Neither part reads record                  | −0.16                                    | noise floor   |

**Table 1.** Where the reward for losing comes from. Each number is how many places earlier the worst team drafts than the fifth-worst, on average, which is what finishing last buys. Through 2026 the lottery drew four picks and handed out the rest in order of record. Each row below the first makes one part ignore record. The two parts work against each other, so the ratios do not add up. A weighted draw pulls the worst teams out of the record-ordered picks, hiding how much those picks are worth.

The league's new format confirms this. 3-2-1 draws all sixteen places, so no pick goes by record, and that change carries its whole gain against tanking. Its ball allocation with a four-pick draw behind it would have left 1.76.

Waitlist COLA is the simplest version that clears all three tests. Dropping a team from fifth-worst to worst gains 1.87 places under the original COLA and 1.32 under the 2026 rule, and exactly zero under Waitlist COLA across 720 seasons. It concedes no more to pool-gaming than today's lottery does. The pattern repeats on 26 seasons of real NBA results.

| Mechanism               | Ranked by record<br><i>draft places; want 0</i> | Ranked by own standard<br><i>draft places; want wide</i> |
|-------------------------|-------------------------------------------------|----------------------------------------------------------|
| NBA lottery, 2019 rules | +1.09 ± 0.12                                    | +1.09 ± 0.12                                             |
| 3-2-1 as adopted        | −1.22 ± 0.25                                    | −1.22 ± 0.25                                             |
| Classic COLA (original) | +1.88 ± 0.11                                    | +1.38 ± 0.24                                             |
| <b>Waitlist COLA</b>    | <b>−0.12 ± 0.22</b>                             | <b>+1.71 ± 0.13</b>                                      |

**Table 2.** The same measurement under two rankings. Rank the fourteen lottery teams, take the first and the fifth, and compare the picks they get. Ranked by record, the number is what losing buys, and zero is the goal. Ranked by the rule's own standard of who deserves help, a big positive number means the rule delivers. When a rule's standard is record, both rankings give the same list, so the two columns must match. A negative number flips the reading, so 3-2-1 gives its worst teams the latest picks. The original COLA fills picks 5 through 14 in order of record, which is the one setting Waitlist COLA changes.

## Conclusion

Ending the reward for losing needs one change. Keep the draw, order the picks behind it by waiting number, and a team gains nothing from losing while the longest-suffering teams still pick earliest, more than the rule the league just adopted manages. Lottery night would look the same to fans. A league weighing these goals differently can move the seven settings and score the result on the same three tests.